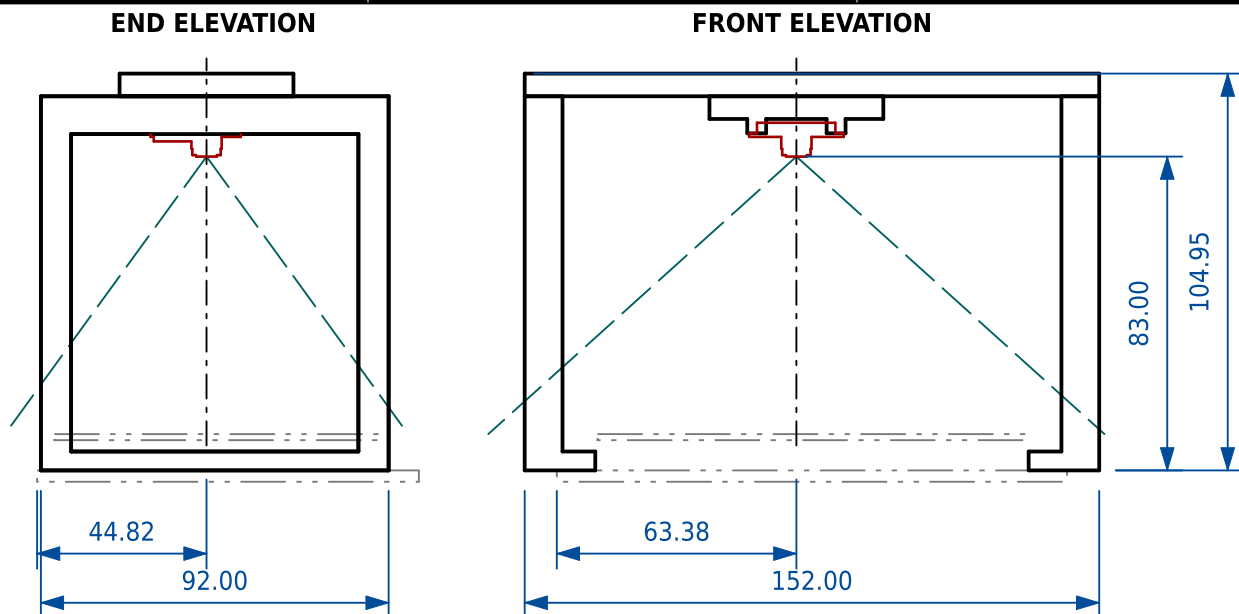
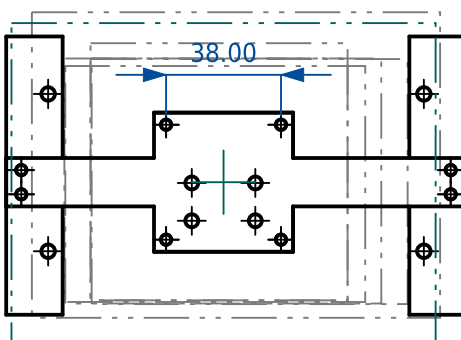




PLAN, 1:2.5, NOT IN PROJECTION



NOTES

1. First angle, in TT-MP-PLATE's coordinates, Z up from the plate's TOP face. The right side frame is the left one mirrored.
2. FOR: the lens over frame A of RPICAM-OVER-PLATE (not drawn there for this lens: optics.place() of frame A and the lens gives the height) at (63.38, 44.82), high enough for the 120 deg lens, 96.00 x 72.00 deg, to take in every revision's whole board: 72.40 over the highest board face, 9.60 up, so 82.00. Set at 83.00, 1.00 over for the print, in a mode reading the whole sensor.
3. THE SAME BEAM AND CARRIER as TT-MP-CAM65, only 67.00 mm lower: only the side frames change with the lens.

NOTES (continued)

4. CAMERA: a v1.3's board with a 120 deg lens ASSUMED on it, face 5.20 off the board; Arducam B006604 itself is a Pi Zero spy camera this carrier does not take. A taller lens lowers the face by the difference, and 1.00 is all there is: measure the lens first.
5. TURN THE CAMERA RIGHT: ASSUMED that the picture's long side runs along the camera board's 25 mm width, which nobody publishes; it has to run along X, and a quarter turn out the picture is 16.8 mm short at each end. If it is, turn the carrier a quarter: its fixings are square about the lens.
6. FOCUS: declared "Focus Distance 1 m to infinity"; at 73.4 the boards are OUT OF FOCUS, a point about 18 pixels, 0.8 mm on the board (RPICAM-OVER-PLATE).
7. MAKE the parts from the STEP files, tt-camera-holder-120-side-left and -right and the shared -beam and -carrier; this sheet is the assembly. Holes: M4 4.5, M3 3.4, M2 2.4, M2 nut pockets 4.3 AF. Ream the side frames' holes, which print lying down. Opaque PETG, four perimeters, matte black if it can be: the side frames show in the picture's margin. The camera's holes are 2.0 +/-0.2: open a tight one with a 2.0 drill.

NOTES (continued)

8. ACCESS: front and back open over every Pmod host and USB-C, the side frames windows over the unfitted side Pmods J12-J14; buttons, not in the board data, are pressed from above, under a beam 99 mm up.
9. FIXING: the feet share the plate's own M4 side fixings, so no new hole; on a chassis the M4s are 5 mm longer than the plate's own. Plug the FFC in first -- its latch is out of reach under the carrier -- and take it up and back over the beam.

SOURCES

1. Where the camera goes: raspberry_pi_camera/optics.py - The plate subject's frame A and the 120 deg lens: RPICAM-OVER-PLATE.
2. What it stands on: tinytapeout/mounting_plate/plate.py - Outline, side fixings, standoff; TT-MP-PLATE.
3. What it carries: raspberry_pi_camera/v1.py - The Camera Module v1.3: Gert van Loo's hand-measured sheet of 21 May 2013, checked against Raspberry Pi Spy's calipers.

PARTS, PRINTED

PART	QTY	X x Y x Z mm	PRINT
Side frame	2	18.7 x 92.0 x 99.0	Print on its outer face
Beam	1	152.0 x 46.0 x 6.0	Print flat, on its top face
Camera carrier	1	46.0 x 46.0 x 9.8	Print on its top face, bosses up

FASTENERS

QTY	SCREW	WITH	JOINS
4	M4 x 16 button head, ISO 7380	washer and nut	Foot to plate
4	M3 x 20 socket head, ISO 4762	and nut	Beam to the side frames' rails
4	M3 x 16 socket head, ISO 4762	and nut	Carrier to the beam's pad
4	M2 x 10 socket head, ISO 4762	and nut	Camera to the carrier

HEIGHTS ABOVE THE PLATE FACE, mm

	Z
Board faces	9.56 to 9.60
Lens face, least	82.00
Lens face, as built	83.00
Camera board	88.20 to 89.15
Carrier	92.95 to 98.95
Beam	98.95 to 104.95

LEGEND

- Printed part
- Camera Module, bought
- The plate and every revision's board, adjacent
- The 120 deg lens's field of view, to its picture's edge
- Dimension, extension and leader
- Frame A of RPICAM-OVER-PLATE, on the board plane

CAMERA HOLDER, 120 DEG LENS

Holds a Camera Module over every demo board revision

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-591-g66eadf0	mm	1:2, PLAN 1:2.5
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	TT-MP-CAM120	A
MATERIAL	GENERAL TOLERANCE	PROJECTION	
PETG, printed, opaque	print +/-0.2, Z DERIVED		